

# My Project

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# Chapter 1

## My Project

### 1.1 Introduction

FlexBox is a flexible MATLAB toolbox for finite dimensional convex variational problems in image processing and beyond. Such problems often consist of non-differentiable parts and involve linear operators. The toolbox uses a primal-dual scheme to avoid (computationally) inefficient operator inversion and to get reliable error estimates. From the user-side, FlexBox expects the primal formulation of the problem, automatically decouples operators and dualizes the problem. For large-scale problems, FlexBox also comes with a C++ module, which can be used stand-alone or together with MATLAB via MEX-interfaces. Besides various pre-implemented data-fidelities and regularization-terms, FlexBox is able to handle arbitrary operators while being easily extendable, due to its object-oriented design. The toolbox is available at <http://www.flexbox.im>

Nowadays, many problems in image processing consist of minimizing convex energies. Typically these problems can be written as  $\min_x F(Ax)$



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